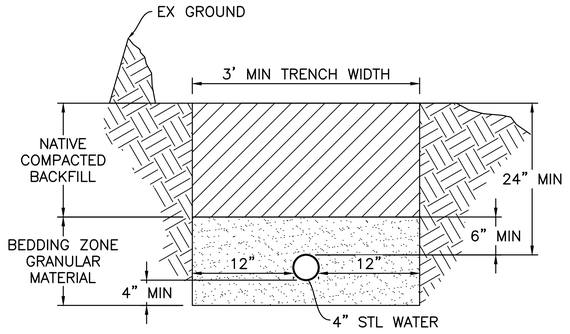
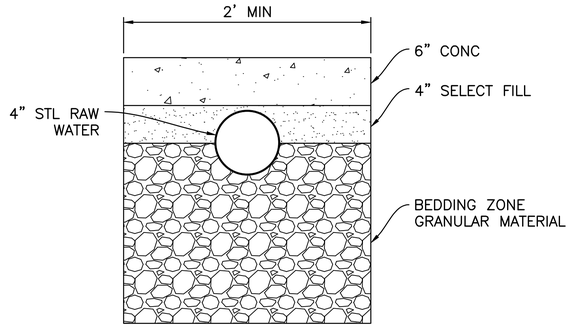


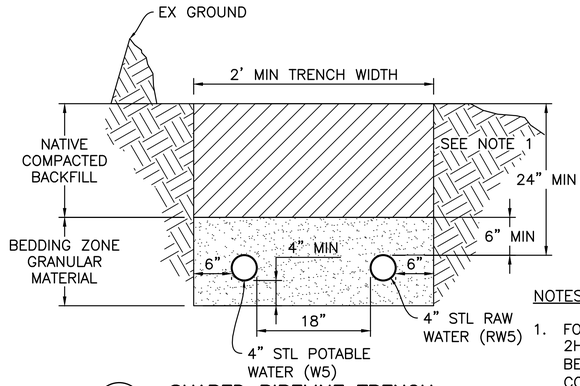
1 051C501 **SINGLE PIPELINE TRENCH DETAIL**

NOT TO SCALE



2 051C501 **TYP TRENCH – PIPELINE BELOW RSPH DECK**

NOT TO SCALE

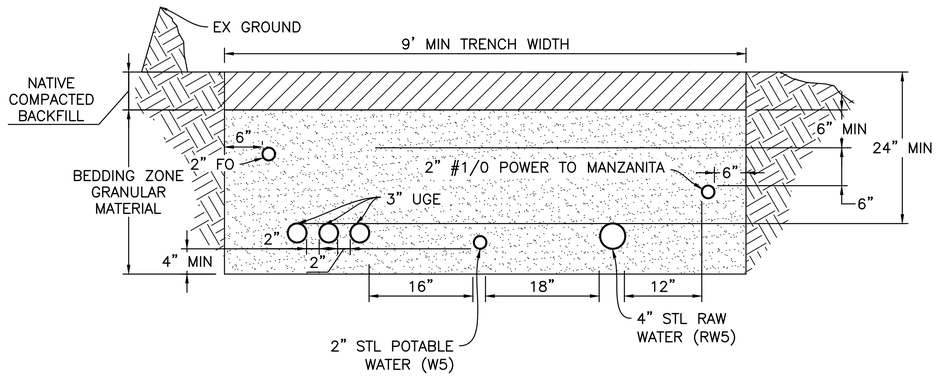


NOTES:

1. FOR SLOPES EXCEEDING 2H:1V, MIN COVER TO BE 30". OTHERWISE, MIN COVER TO BE 24".

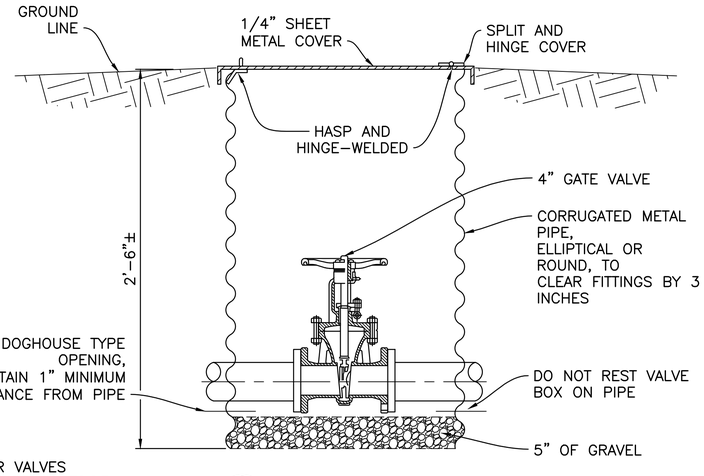
3 051C501 **SHARED PIPELINE TRENCH**

NOT TO SCALE



4 051C501 **SHARED UTILITY TRENCH**

NOT TO SCALE

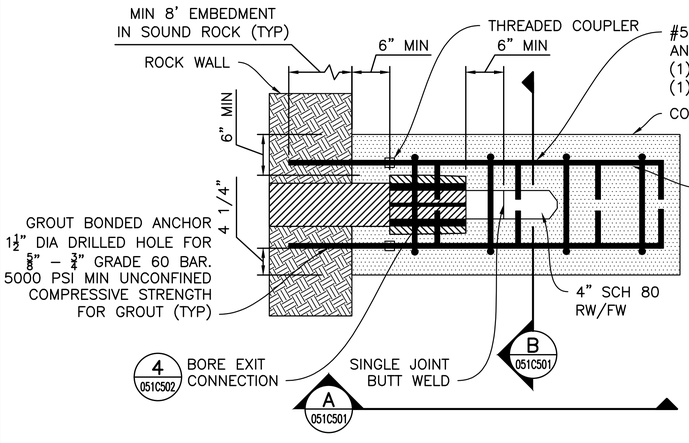


NOTE:

1. FOR VALVES LOCATED OUTSIDE OF TRAIL

5 051C501 **CMP VALVE BOX**

NOT TO SCALE

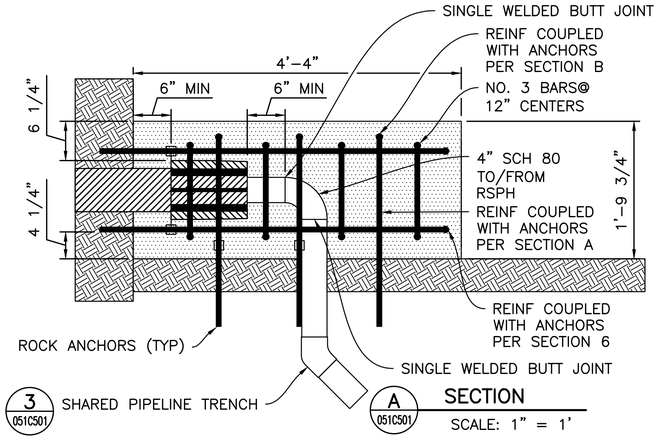


NOTES:

1. MINOR DEVIATION IN ENCASEMENT DIMENSIONS BASE ON EXISTING GROUND CONDITION ARE ANTICIPATED TO FACILITATE CONSTRUCTION.
2. IN LIEU OF COUPLER AND HOOKED BAR, ANCHORS MAY BE EXTENDED AND HOOKED OR HEADED AS APPROVED BY MANUFACTURER.

6 051C501 **WATER LINE CONCRETE ENCASEMENT AT BORE EXIT**

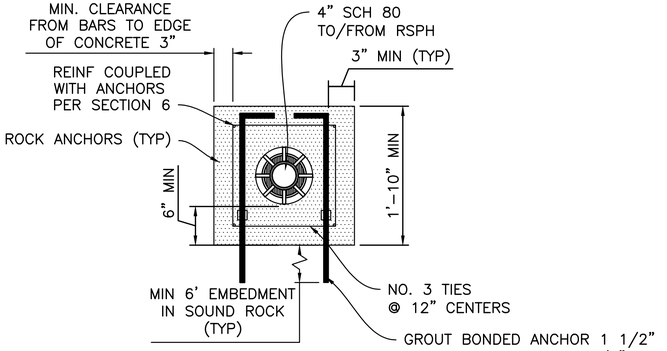
SCALE: 1" = 1'



3 051C501 SHARED PIPELINE TRENCH

A 051C501 **SECTION**

SCALE: 1" = 1'



B 051C501 **SECTION**

SCALE: 1" = 1'

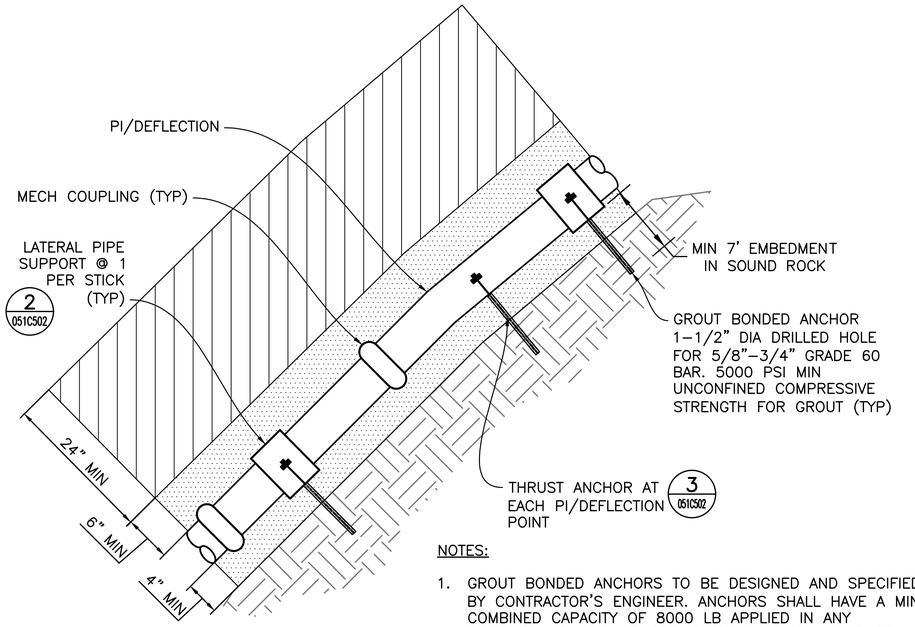
Registered Professional Engineer (Civil) CERTIFICATE NO. 22398 NICHOLAS J. LAFRONZ Date Signed 10-9-2026 ARIZONA, U.S.A.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY Date Signed 10-9-2026 ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE ADD C. WITTENMEIER TECH. REVIEW: GOUGH/GLOVER DATE: JUNE 2026	SUB SHEET NO. 051 C501	TITLE OF SHEET SITEWORK CIVIL DETAILS TRENCHING NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 21 OF 113

10/13/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\051-C501.dwg



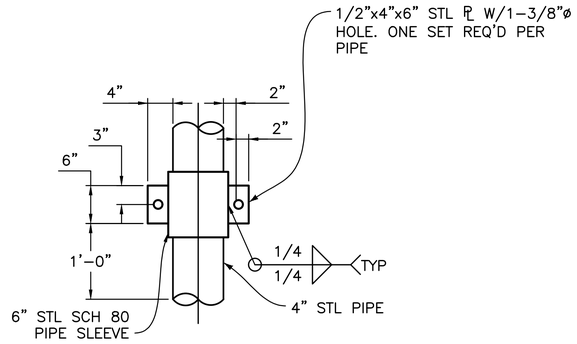
TYPICAL PROFILE

NOTES:

1. GROUT BONDED ANCHORS TO BE DESIGNED AND SPECIFIED BY CONTRACTOR'S ENGINEER. ANCHORS SHALL HAVE A MIN COMBINED CAPACITY OF 8000 LB APPLIED IN ANY DIRECTION ON ANCHOR GROUP AND INDIVIDUAL ANCHOR CAPACITY OF 4000 LB SHEAR AND 4000 LB TENSION.
2. ALL STEEL HARDWARE SHALL BE GALVANIZED GRADE 60 UNLESS OTHERWISE NOTED.

1 051C502 TYPICAL PIPE PROFILE WITH MECHANICAL COUPLINGS LATERAL THRUST ANCHORS AND PIPE SUPPORTS

NOT TO SCALE

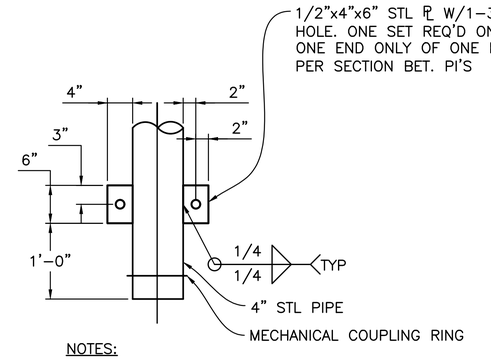


NOTES:

1. LATERAL PIPE SUPPORTS TO BE PLACED (1) AT EVERY STICK OF PIPE WITH MECHANICAL COUPLINGS.
2. REFER TO DETAIL 1/051C502 FOR SUPPORT LOCATIONS.

2 051C502 MECHANICAL COUPLING TYPICAL LATERAL PIPE SUPPORT PLAN

NOT TO SCALE

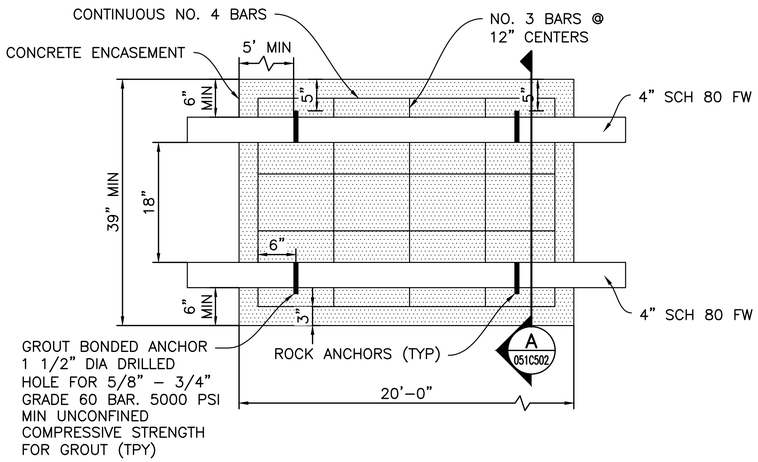


NOTES:

1. PIPE THRUST SUPPORTS TO BE PLACED AT EVERY DEFLECTION POINT/POINT OF INFLECTION.
2. REFER TO DETAIL 1/051C502 FOR SUPPORT LOCATIONS.

3 051C502 MECHANICAL COUPLING PIPE THRUST ANCHOR PLAN

NOT TO SCALE

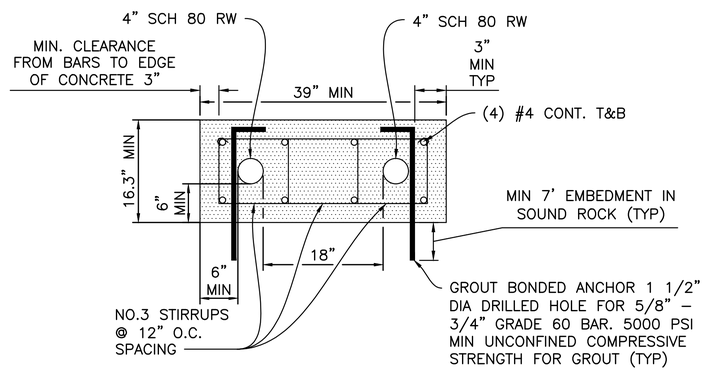


4 051C502 WATER LINE CONCRETE ENCASEMENT AT DRAINAGE CROSSING

NOT TO SCALE

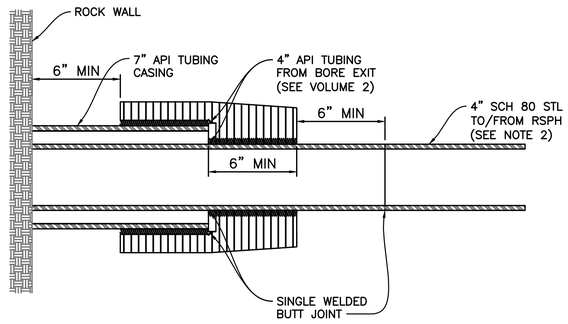
NOTES:

1. MINOR DEVIATION IN ENCASEMENT DIMENSIONS BASED ON EXISTING GROUND CONDITION ARE ANTICIPATED TO FACILITATE CONSTRUCTION.



A 051C502 SECTION

NOT TO SCALE



5 051C502 BORE EXIT CONNECTION

NOT TO SCALE

NOTES:

1. DO NOT TORQUE BOLTS ON 900# FLANGES MORE THAN 2 TIMES BEFORE REPLACING HARDWARE.
2. CONTRACTOR TO MAKE FIELD ADJUSTMENTS BASED ON CONSTRUCTED BORE EXIT LOCATIONS.

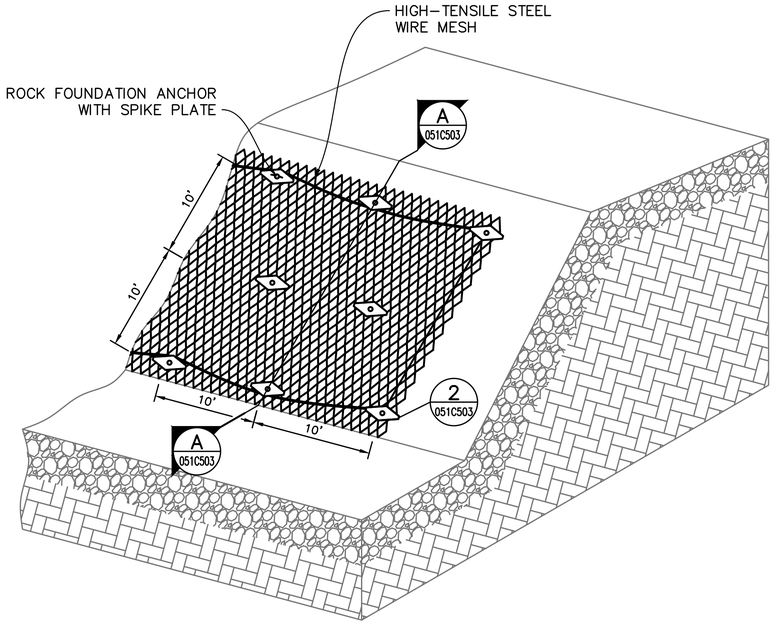
Registered Professional Engineer CERTIFICATE NO. 22398 NICHOLAS J. LAFRONZ Date Signed 10-9-26 ARIZONA, U.S.A.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY Date Signed 10-9-2026 ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/GLOVER DATE: JUNE 2026	SUB SHEET NO. 051 C502	TITLE OF SHEET SITEWORK CIVIL DETAILS PIPE ANCHORS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 22 OF 113

10/12/2026 Cowles, Byron C:\Users\bcowles\DC\ACCDocs\HDR\10414450_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAD_HDR\DRAWINGS\CIVIL\051-C502.dwg

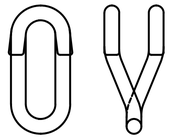




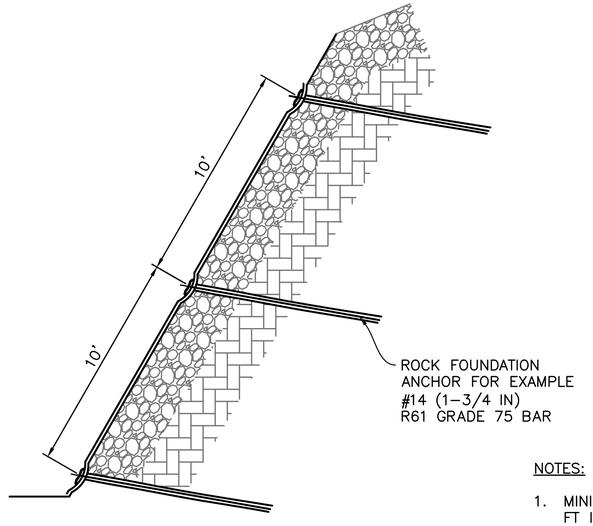
SPIKE PLATE



CONNECTION
CLIP



PRESS CLAW

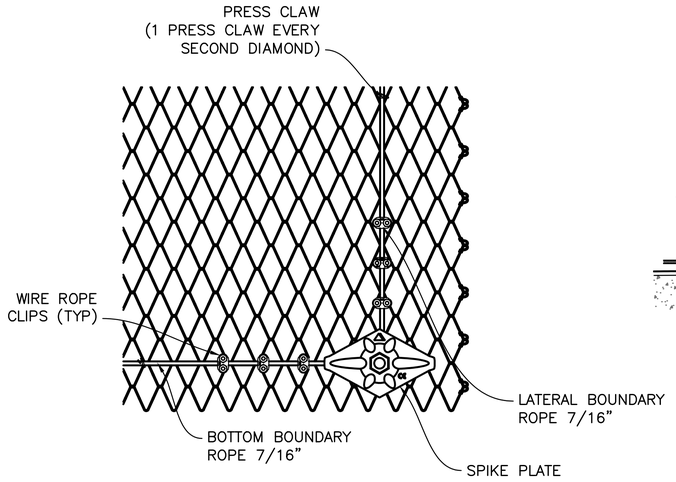


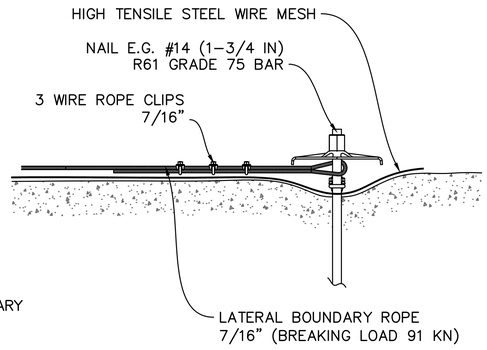
NOTES:

1. MINIMUM EMBEDMENT LENGTH OF 10 FT INTO COMPETENT UNDISTURBED ROCK. REQUIRED EMBEDMENT TO BE DETERMINED IN ACCORDANCE WITH SPECIFICATION SECTIONS 31 32 19 AND 31 68 13.

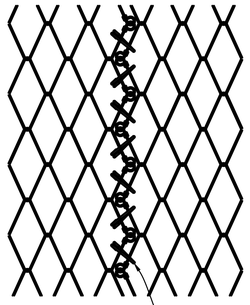
1 / 051C503 GENERAL ROCK FOUNDATION ANCHOR ARRANGEMENT DETAIL
NOT TO SCALE

AA / 051C503 SECTION
NOT TO SCALE





2 / 051C503 WIRE MESH BOUNDARY CONDITION – SHEAR FORCE ON THE NAIL
NOT TO SCALE



3 / 051C503 HIGH TENSILE STEEL WIRE MESH CONNECTION – TYP. VERTICAL WITHOUT OVERLAP
NOT TO SCALE

Registered Professional Engineer
CERTIFICATE NO. 22198
NICOLAS D. LAFRONZ
Date Signed 5-1-2026
ARIZONA, U.S.A.

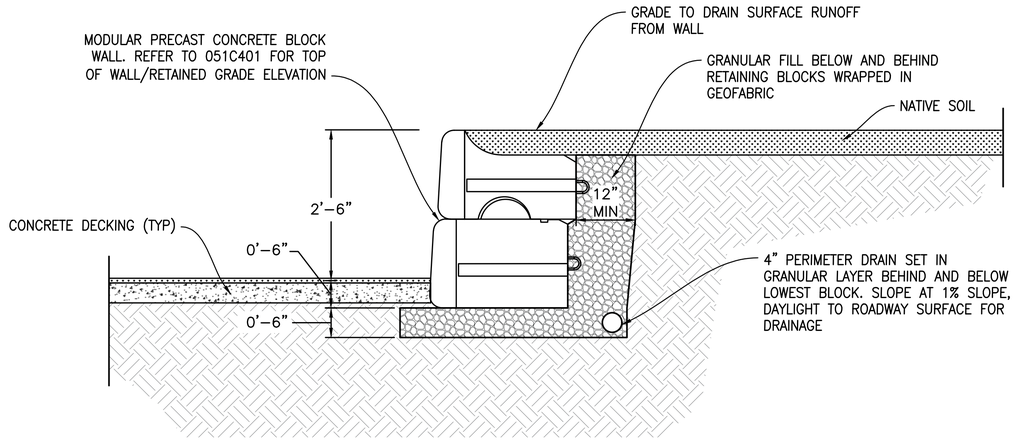
Registered Professional Engineer (Civil)
CERTIFICATE NO. 76139
NICHOLAS HUMPHREY
5-1-2026
Date Signed
ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD C. WITTENMEIER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 051 C503	TITLE OF SHEET SITEWORK CIVIL DETAILS HIGH TENSILE WIRE MESH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 23 OF 113

4/17/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\051-C503.dwg

NOTES:

1. REFER TO SPECIFICATION 32 32 16 – PRECAST MODULAR BLOCK RETAINING WALL FOR MODULAR BLOCK WALL DESIGN AND CONSTRUCTION.



1 / 051C401 MODULAR WALL DETAIL

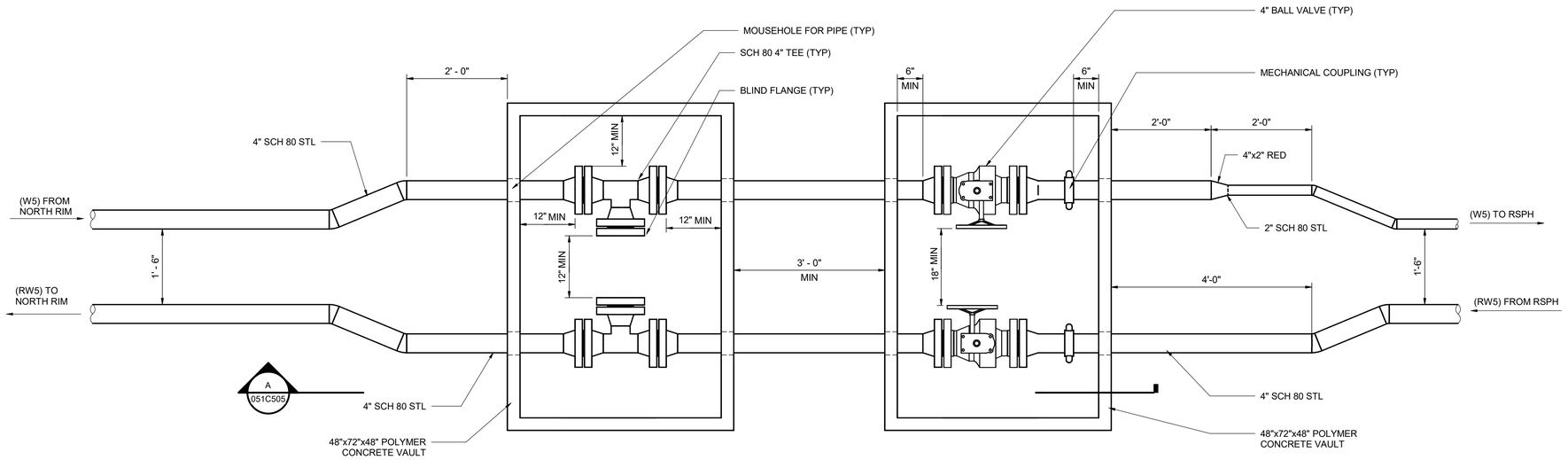
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

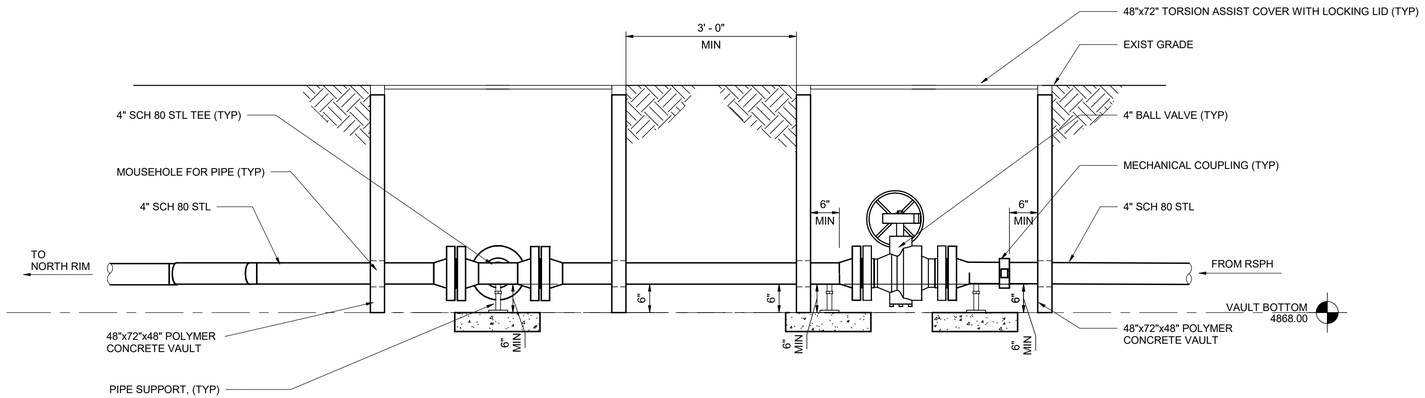
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD C. WITTENMEIER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 051 C504	TITLE OF SHEET SITEWORK CIVIL DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 24 OF 113

4/17/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\051-C504.dwg



ISOLATION VALVE VAULT PLAN
SCALE (A)



A 051C505 SECTION
SCALE (A)

GENERAL NOTES:

A. FLANGED CONNECTIONS SHOWN ON DRAWINGS ARE TO BE ANSI CLASS 900.

B. DO NOT TORQUE BOLTS ON 900# FLANGES MORE THAN TWO TIMES BEFORE REPLACING HARDWARE.

SCALE (A) 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: AUG 2025	SUB SHEET NO. 051 C505	TITLE OF SHEET SITEWORK ISOLATION VALVE VAULT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 25 OF 113

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4/27/2026 5:38:54 PM